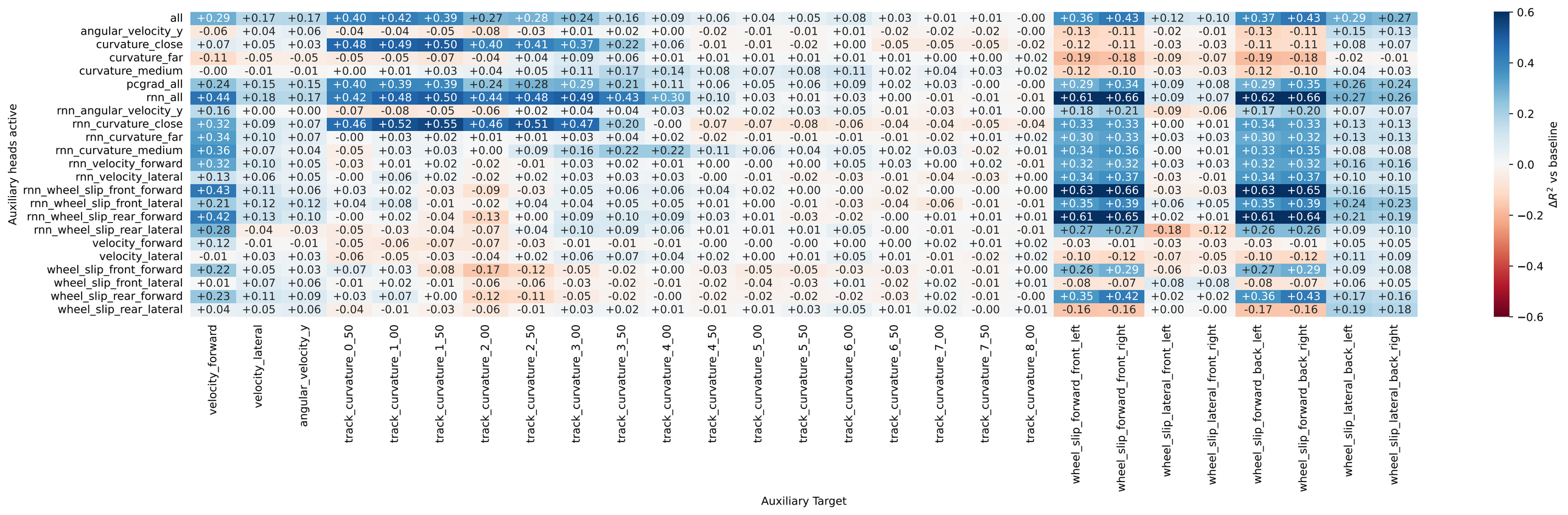
Auxiliary heads active
all +0.29 +0.17 +0.17 +0.40 +0.42 +0.39 +0.27 +0.28 +0.24 +0.16 +0.09 +0.06 +0.04 +0.05 +0.08 +0.03 +0.01 +0.01 -0.00 +0.36 +0.43 +0.12 +0.10 +0.37 +0.43 +0.29 +0.27
angular_velocity_y -0.06 +0.04 +0.06 -0.04 -0.04 -0.05 -0.08 -0.03 +0.01 +0.02 +0.00 -0.02 -0.01 -0.01 +0.01 -0.02 -0.02 -0.02 -0.00 -0.13 -0.11 -0.02 -0.01 -0.13 -0.11 +0.15 +0.13
curvature_close +0.07 +0.05 +0.03 +0.48 +0.49 +0.50 +0.40 +0.41 +0.37 +0.22 +0.06 -0.01 -0.01 -0.02 +0.00 -0.05 -0.05 -0.05 -0.02 -0.12 -0.11 -0.03 -0.03 -0.11 -0.11 +0.08 +0.07
curvature_far -0.11 -0.05 -0.05 -0.05 -0.05 -0.07 -0.04 +0.04 +0.09 +0.06 +0.01 +0.01 +0.01 +0.01 +0.01 +0.01 +0.00 +0.00 +0.02 -0.19 -0.18 -0.09 -0.07 -0.19 -0.18 -0.02 -0.01
curvature_medium -0.00 -0.01 -0.01 +0.00 +0.01 +0.03 +0.04 +0.05 +0.11 +0.17 +0.14 +0.08 +0.07 +0.08 +0.11 +0.02 +0.04 +0.03 +0.02 -0.12 -0.10 -0.03 -0.03 -0.12 -0.10 +0.04 +0.03
pcgrad_all +0.24 +0.15 +0.15 +0.40 +0.39 +0.39 +0.24 +0.28 +0.29 +0.21 +0.11 +0.06 +0.05 +0.06 +0.09 +0.02 +0.03 -0.00 -0.00 +0.29 +0.34 +0.09 +0.08 +0.29 +0.35 +0.26 +0.24
rnn_all +0.44 +0.18 +0.17 +0.42 +0.48 +0.50 +0.44 +0.48 +0.49 +0.43 +0.30 +0.10 +0.03 +0.01 +0.03 +0.00 -0.01 -0.01 -0.01 +0.61 +0.66 +0.09 +0.07 +0.62 +0.66 +0.27 +0.26
rnn_angular_velocity_y +0.16 +0.00 +0.00 -0.07 -0.08 -0.05 -0.06 +0.02 +0.04 +0.04 +0.03 +0.02 +0.02 +0.03 +0.05 -0.01 -0.03 +0.01 -0.00 +0.18 +0.21 -0.09 -0.06 +0.17 +0.20 +0.07 +0.07
rnn_curvature_close +0.32 +0.09 +0.07 +0.46 +0.52 +0.55 +0.46 +0.51 +0.47 +0.20 -0.00 -0.07 -0.07 -0.08 -0.06 -0.04 -0.04 -0.05 -0.04 +0.33 +0.33 +0.00 +0.01 +0.34 +0.33 +0.13 +0.13
rnn_curvature_far +0.34 +0.10 +0.07 -0.00 +0.03 +0.02 +0.01 +0.01 +0.03 +0.04 +0.02 -0.02 -0.01 -0.01 -0.01 -0.01 -0.02 +0.01 +0.02 +0.30 +0.33 +0.03 +0.03 +0.30 +0.32 +0.13 +0.13
rnn_curvature_medium +0.36 +0.07 +0.04 -0.05 +0.03 +0.03 +0.00 +0.09 +0.16 +0.22 +0.22 +0.11 +0.06 +0.04 +0.05 +0.02 +0.03 -0.02 +0.01 +0.34 +0.36 -0.00 +0.01 +0.33 +0.35 +0.08 +0.08
rnn_velocity_forward +0.32 +0.10 +0.05 -0.03 +0.01 +0.02 -0.02 -0.01 +0.03 +0.02 +0.01 +0.00 -0.00 +0.00 +0.05 +0.03 +0.00 +0.02 -0.01 +0.32 +0.32 +0.03 +0.03 +0.32 +0.32 +0.16 +0.16
rnn_velocity_lateral +0.13 +0.06 +0.05 -0.00 +0.06 +0.02 -0.02 +0.02 +0.03 +0.03 +0.03 -0.00 -0.01 -0.02 -0.03 -0.01 -0.04 -0.03 +0.00 +0.34 +0.37 -0.03 -0.01 +0.34 +0.37 +0.10 +0.10
rnn_wheel_slip_front_forward +0.43 +0.11 +0.06 +0.03 +0.02 -0.03 -0.09 -0.03 +0.05 +0.06 +0.06 +0.04 +0.02 +0.00 -0.00 -0.02 -0.00 -0.00 +0.00 +0.63 +0.66 -0.03 -0.03 +0.63 +0.65 +0.16 +0.15
rnn_wheel_slip_front_lateral +0.21 +0.12 +0.12 +0.04 +0.08 -0.01 -0.02 +0.04 +0.04 +0.05 +0.05 +0.01 +0.00 -0.01 -0.03 -0.04 -0.06 -0.01 -0.01 +0.35 +0.39 +0.06 +0.05 +0.35 +0.39 +0.24 +0.23
rnn_wheel_slip_rear_forward +0.42 +0.13 +0.10 -0.00 +0.02 -0.04 -0.13 +0.00 +0.09 +0.10 +0.09 +0.03 +0.01 -0.03 -0.02 -0.00 +0.01 +0.01 +0.01 +0.61 +0.65 +0.02 +0.01 +0.61 +0.64 +0.21 +0.19
rnn_wheel_slip_rear_lateral +0.28 -0.04 -0.03 -0.05 -0.03 -0.04 -0.07 +0.04 +0.10 +0.09 +0.06 +0.01 +0.01 +0.01 +0.05 -0.00 -0.01 -0.01 -0.01 +0.27 +0.27 -0.18 -0.12 +0.26 +0.26 +0.09 +0.10
velocity_forward +0.12 -0.01 -0.01 -0.05 -0.06 -0.07 -0.07 -0.03 -0.01 -0.01 -0.01 -0.00 -0.00 -0.00 +0.00 +0.00 +0.02 +0.01 +0.02 -0.03 -0.01 -0.03 -0.03 -0.03 -0.01 +0.05 +0.05
velocity_lateral -0.01 +0.03 +0.03 -0.06 -0.05 -0.03 -0.04 +0.02 +0.06 +0.07 +0.04 +0.02 +0.00 +0.01 +0.05 +0.01 -0.01 -0.02 +0.02 -0.10 -0.12 -0.07 -0.05 -0.10 -0.12 +0.11 +0.09
wheel_slip_front_forward +0.22 +0.05 +0.03 +0.07 +0.03 -0.08 -0.17 -0.12 -0.05 -0.02 +0.00 -0.03 -0.05 -0.05 -0.03 -0.03 -0.01 -0.01 +0.00 +0.26 +0.29 -0.06 -0.03 +0.27 +0.29 +0.09 +0.08
wheel_slip_front_lateral +0.01 +0.07 +0.06 -0.01 +0.02 -0.01 -0.06 -0.06 -0.03 -0.02 -0.01 -0.02 -0.04 -0.03 +0.01 -0.02 +0.02 -0.01 +0.01 -0.08 -0.07 +0.08 +0.08 -0.08 -0.07 +0.06 +0.05
wheel_slip_rear_forward +0.23 +0.11 +0.09 +0.03 +0.07 +0.00 -0.12 -0.11 -0.05 -0.02 -0.00 -0.02 -0.02 -0.02 -0.02 -0.03 +0.02 -0.01 -0.00 +0.35 +0.42 +0.02 +0.02 +0.36 +0.43 +0.17 +0.16
wheel_slip_rear_lateral +0.04 +0.05 +0.06 -0.04 -0.01 -0.03 -0.06 -0.01 +0.03 +0.02 +0.01 -0.01 +0.01 +0.03 +0.05 +0.01 +0.02 -0.00 +0.01 -0.16 -0.16 +0.00 -0.00 -0.17 -0.16 +0.19 +0.18
velocity_forward
velocity_lateral
angular_velocity_y
track_curvature_0_50
track_curvature_1_00
track_curvature_1_50
track_curvature_2_00
track_curvature_2_50
track_curvature_3_00
track_curvature_3_50
track_curvature_4_00
track_curvature_4_50
track_curvature_5_00
track_curvature_5_50
track_curvature_6_00
track_curvature_6_50
track_curvature_7_00
track_curvature_7_50
track_curvature_8_00
wheel_slip_forward_front_left
wheel_slip_forward_front_right
wheel_slip_lateral_front_left
wheel_slip_lateral_front_right
wheel_slip_forward_back_left
wheel_slip_forward_back_right
wheel_slip_lateral_back_left
wheel_slip_lateral_back_right
Auxiliary Target
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
ΔR^2 vs baseline